Wound Drainage Systems Procedure

1. Scope

Site	Operational Area	Applicable to
Armadale Kalamunda Group	All clinical services	Nursing and Midwifery

2. Purpose

The purpose of this document is to establish minimum practice standard for wound drainage systems throughout Armadale Kalamunda Group (AKG).

3. Closed drainage system

3.1 Changing a closed drainage system canister

Drainage canisters should only be changed when the drainage canister is full; the vacuum is running out, or on the instruction of the Medical Officer (MO). Do not attempt to re-vacuum a closed drainage system.

3.2 Equipment Required

- New Sterile Vacuum drainage system
- PPE
- Alcohol Swabs (70% Alcohol, 2% Chlorhexidine)
- Rubbish bag

3.3 Procedure for changing a closed drainage system canister

- Ensure privacy and explain procedure to patient
- Assemble required equipment and confirm drainage required as per post-operative instructions/current MO instructions
- Perform hand hygiene, maintain aseptic and/or aseptic non- touch technique (ANTT) and apply PPE
- Clamp off evacuation tubing close to the canister to be changed
- Clamp the patient's evacuation tube
- Clean around connection site on the evacuation tube from the canister with alcohol swabs for 20 seconds and allow to air dry (30 seconds)
- Open packaging of new drainage canister and check that it is clamped and vacuumed
- Using non touch technique, disconnect drainage tube from canister to be changed and reconnect drainage tube to new canister
- Release the clamp on the canister and patient's evacuation tubing
- Varivac- Ensure dial is set as per post-operative instructions/current MO instructions
- Privac- Ensure vacuum indicator is compressed

- Dispose of all items correctly and safely (clinical waste/sharps container), perform hand hygiene
- Record date and time canister changed, and drainage measured on patient's health care records and any other required documentation, (fluid balance or observations charts)

If there is loss of vacuum after changing canister check site and position of surgical drain.

3.4 Removal of a closed drainage system

Drains are only to be removed on instruction from MO. Ensure instruction to remove drain is documented by the MO in patient's health care record. The removal of a vacuum sealed drain can be painful, so assessment of patient should be made, and appropriate analgesia should be prescribed and administered prior to procedure being undertaken.

3.5 Equipment required

- Sterile dressing pack
- Sterile gauze swabs
- PPE
- Sodium Chloride 0.9%
- Absorbent dressing
- Hypoallergenic adhesive tape
- Stitch cutter
- Non-sterile gloves
- Rubbish bag

3.6 Procedure for removal of a closed drainage system

- Ensure privacy and explain procedure to patient
- Perform and document pain score and administer analgesia prior to procedure as prescribed by MO and allow to take effect
- Varicac- Release vacuum 5 minutes prior to removal by dialing canister vacuum to gravity. Following this, clamp canister and drainage tube.
- Privac- Clamp evacuation tube on drain canister; ensure clamp on drain tubing is left open, as per manufacturer's instructions.
- Assemble required equipment
- Assist patient into a comfortable position with drain site accessible
- Perform hand hygiene, maintain aseptic and/or aseptic non- touch technique (ANTT) and apply PPE
- Remove existing dressing, ensure drain remains in place
- Remove and discard gloves, perform hand hygiene and don clean gloves
- Cleanse drain insertion site with sodium chloride 0.9%
- If drain held in place by suture, remove suture with stitch cutter
- Support skin at drain insertion site with gauze
- Using forceps gently withdraw the drain
- If resistance is encountered, stop procedure, re-dress area with sterile dressing, contact MO

- Once drain removed, apply pressure using gauze
- Clean wound site and apply appropriate dressing
- Ensure patient is comfortable
- Dispose of all items correctly and safely (clinical waste/sharps container), perform hand hygiene
- Record date and time of drain removal and drainage measured on patient's health care records and any other required documentation, (fluid balance or observations charts).

4. Removal of open drainage system

Drainage systems are only to be removed on instruction from MO. Ensure instruction to remove drainage system is documented by the MO in patient's health care record.

4.1 Equipment

- Sterile dressing Pack
- PPE
- Sodium Chloride0.9%
- Non-sterile gloves
- Stitch cutter
- Sterile forceps
- Absorbent dressing
- Hypoallergenic adhesive tape
- Rubbish bag

4.2 Procedure for removal of open drainage system

- Ensure privacy and explain procedure to patient
- Perform and document pain score and administer analgesia prior to procedure as prescribed by MO and allow to take effect
- Assemble required equipment
- Assist patient into a comfortable position with drain site accessible
- Perform hand hygiene, maintain aseptic and/or aseptic non- touch technique (ANTT) and don PPE
- Remove dressing from drain insertion site, ensuring drain remains in place
- Remove and discard soiled gloves, perform hand hygiene and don clean gloves
- Cleanse area around drain insertion site with sodium chloride 0.9%
- If drain held in place by suture, remove suture with stitch cutter as close to skin as possible
- Support skin at drain insertion site with gauze
- Using forceps and gauze gently withdraw the drain
- If resistance is encountered, stop procedure, re-dress area with sterile dressing, contact MO
- Once drain is removed, clean the wound site, and apply appropriate dressing

5. Shortening of open drainage system

Drainage systems are only to be shortened on instruction from MO. Ensure instruction to shorten drainage system is documented by the MO in patient's health care record. To prevent the drain slipping into the wound, always insert sterile safety pin into proximal part prior to shortening and cutting drain.

5.1 Equipment

- Sterile dressing Pack
- PPE
- Sodium Chloride0.9%
- Sterile gauze swabs
- Sterile Gloves
- Non-sterile gloves
- Stitch cutter
- Sterile scissors
- Sterile forceps
- Sterile safety pin.
- Sterile dressing or drainage bag
- Hypoallergenic adhesive tape
- Rubbish bag

5.2 Procedure for shortening of open drainage system

- Ensure privacy and explain procedure to patient
- Perform and document pain score and administer analgesia prior to procedure as prescribed by MO
- Assemble required equipment
- Assist patient into a comfortable position with drain site accessible
- Perform hand hygiene, maintain aseptic and/or aseptic non- touch technique (ANTT) and don PPE
- Remove dressing from drain insertion site, ensuring drain remains in place
- Remove and discard soiled gloves, perform hand hygiene, and don clean gloves
- Cleanse area around drain insertion site with sodium chloride 0.9%
- Place sterile safety pin through the drain as close to skin as possible.
- If drain held in place by suture, remove suture with stitch cutter as close to skin as possible
- Support skin at drain insertion site with gauze
- Remove gloves.
- Apply sterile gloves and gently ease the drain out of the wound to the length documented by MO in the patient's medical record.
- Following shortening of the drainage system, apply a sterile dressing or drainage bag as clinically indicated and appropriate.

- Record date and time of procedure and drainage measured in the patient's medical record and any other required documentation

6. Documentation

Check drains during handover and during hourly rounding.

Document fluid drained on Fluid Balance Chart when changing or removing drain.

Ensure at midday (1200hrs) and at midnight (2400hrs) the drain is marked at the fluid level at that time. Keep a cumulative total of fluid drained on Fluid Balance Chart.

Failure to record accurately and legibly, and understand what is recorded, in patient health records contribute to a decrease in the quality and safety of patient care.

Instructions to shorten, change or remove drainage systems must be documented in the patient's health care record by the MO.

Staff performing all procedures to ensure documentation in patient health care record.

7. Legislative context

The following documents are required to give effect to this procedure:

- Health Department of Western Australia (HDWA) Clinical Handover Policy MP 0095
- East Metropolitan Health Service (EMHS) Patient Identification and Procedure Matching Policy EMHS: 21

8. Supporting information

The following documents support the implementation of this procedure:

- Aseptic Technique Procedure AKG-PRO-0045-14
- Hand Hygiene Policy AKG-POL-0187-17
- Pain Management Procedure AKG-PRO-0339-17
- EMHS Consent Policy EMHS: 5

9. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the relevant departmental meeting for review and identification of any opportunities for improvement.

10. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 2 – Action 2.3 Healthcare rights and informed consent
- Standard 3 – Action 3.11 Aseptic technique
- Standard 6 – Action 6.5 Correct identification and procedure matching
- Standard 6 – Action 6.7 Clinical handover

11. Glossary

The following definitions are relevant to this policy.

Term	Definition
Wound draining system	Drainage systems are used to remove body fluid collection following surgery, preventing the accumulation of fluid, improving wound healing, and reducing pain. Drains are classified as either open/passive (non-vacuum) or closed/active (vacuum) depending on their function. A **closed drain** consists of a drainage tube that is placed within a wound and connected to a drainage canister; these are low pressure suction devices that continuously remove fluids against gravity via a vacuum or wall suction. An **open drain** consists of an artificial conduit/tube which is left in a wound to allow fluid drainage and relies on gravity to evacuate the fluid.

12. Policy owner (relating to these procedures)

Director Nursing and Midwifery

Enquiries relating to this procedure may be directed to:

Title: Coordinator of Nursing
Department: Surgical Services
Email: shane.hastings@health.wa.gov.au

13. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4	31/01/2024	31/01/2027	Scheduled review

14. Authorisation

This procedure has been authorised and issued by the Director Nursing and Midwifery.

Authorised by	Denver Prince – A/Director Nursing and Midwifery
Authorisation date	30/01/2024
Published date	31/01/2024

Appendix 1: Most commonly used closed drainage system

1. Varivac™

1.1 Changing a canister

- Prior to clamping turn dial to gravity
- Disconnect Varivac™ drainage canister at luer lock connector
- Following attachment of new drainage canister ensure the dial is set as per post-operative instructions/current MO requirements.
- The vacuum range is infinitely adjustable between the approximate settings below:
    - Gravity 0kPa
    - Low 5kPa (37mmHg)
    - Medium 10kPa (75mmHg)
    - High 15kPa (113mmHg)

1.2 Removing a drainage system

- Prior to removal of Varivac™ drainage system turn the suction dial on the drainage canister to gravity and leave for 5 minutes to release suction from the drain
- Following this, clamp canister and drainage tube
- Do not disconnect Varivac™ drainage canister from drain tubing, dispose of whole system in clinical waste.

2. Privac™

2.1 Changing a canister

- Ensure vacuum indicator is compressed

2.2 Removing a drainage system

- Clamp evacuation tube on drainage canister; ensure clamp on drain tubing is left open.
- Disconnect canister from evacuation tubing at luer lock to release vacuum – care is required to prevent any release of body fluids
- Reconnect drainage canister to evacuation tubing at luer lock

3. Survac™

3.1 Changing a canister

- Clamp off tubing close to drainage canister, using attached white tubing clamp
- Clamp the green top of the bottle using the attached white tubing clamp
- Check new drainage canister is clamped and vacuumed
- Following attachment of new drainage canister release the white clamp at the green top on the bottle and then release the tubing clamp